Shivansh

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Education

B.Tech (**Software Engineering**)	Expected 2028	Delhi Technological University, Delhi	7.6 CGPA
CBSE (Class XII)	2023	VCG SBV School	80.8%
CBSE (Class X)	2021	DAV Model School	88.6%

Work Experience

SDE Intern Jun' 25 – Jul' 25
Blue Star Publications *Delhi, India*

- Built a secure admin dashboard (CRUD, JWT auth, role-based access control) exposing **idempotent MERN APIs engineered for reliability**, safely handling concurrent write requests without data corruption.
- Worked with a 5-member team across 2 sprints to gather requirements and **architect the database schema and design patterns** for a legacy codebase, debugging issues and shipping a new feature that improved user engagement.
- Applied data structures and algorithmic reasoning to optimize query paths and reduce redundant reads on high-traffic dashboard endpoints, improving efficiency at scale.

Research Intern – ML/Software Engineering Dec 2025 – Aug 2026
IIT Mandi, CAIR Lab *Mandi, India*

- Engineered an end-to-end PyTorch training pipeline for a large-scale ECG foundation model, processing a **45,152-patient dataset** (PhysioNet Ningbo) with a custom adversarial self-supervised objective (ASGMA + Gradient Reversal Layer), outperforming random-masking baselines.
- Designed and implemented a Transformer architecture with Adaptive Shared Attention (ASA), reaching F1 0.73/AUROC 0.94 (CPSC2018) and F1 0.69/AUROC 0.92 (PTB-XL) – surpassing SimCLR, CMSC, and 3KG baselines.
- Extended the pipeline to a second modality (EEG, Transformer + CBAM with contrastive augmentation), sustaining strong performance at <10% labeled data through label-efficient system design.

Data Analyst Intern – Student Records & Reporting Mar 2026 – May 2026
DTU Administration *Delhi, India*

- Cleaned and standardized 10,000+ student records and automated recurring reporting workflows via formula-driven pipelines (VLOOKUP, QUERY).

Projects

AI-Powered VC Research Automation Engine github.com/5VN5H/Eximius_VC_RA

- *FastAPI, Celery, Redis, PostgreSQL, Claude API, n8n*
- Architected a **distributed, asynchronous data pipeline** (FastAPI + Celery workers + Redis broker) that reliably collects, deduplicates, and structures data **in parallel across multiple enterprise APIs** (Perplexity, Harmonic, AlphaSense), handling concurrent long-running jobs at scale.
- Built a Safe-Write CRM layer (Claude + Pydantic) with PostgreSQL-based audit trails to append AI-generated insights **without overwriting concurrently-edited human data**.

SmartCourt – AI Football Tactical Analysis

- *PyTorch, GNN, GAT, CVAE, StatsBomb 360*
- Built a GNN + Graph Attention Transformer with CVAE-based counterfactual simulation on StatsBomb 360 tracking data, **processing a full match end-to-end in under 3 minutes**; Finalist, TechFest MAIT 2025.

Technical Skills

Languages: C++, Python, JavaScript, SQL
Core CS Fundamentals: Object-Oriented Programming (OOP); Data Structures & Algorithms
Systems & Concurrency: Multi-threading, distributed task queues (Celery/Redis), concurrent request handling, idempotent API design for reliability
Backend & Data Engineering: FastAPI, Node.js/Express, PostgreSQL, SQLAlchemy, MongoDB, REST APIs, JWT Auth, Git, Linux
ML/AI: PyTorch, Scikit-learn, Transformers, GNN, GAT, CVAE

Leadership & Achievements

Co-Head, Paryavarnam – DTU Environmental Society Sep 2025 – Present

- Led a 20-member team for DTU's sustainability drive (2,000+ participants, 3 colleges) and a campus waste-audit across 10+ zones; findings submitted to DTU administration.
- Conducted AI/ML workshops for 200+ students (Classes 10–12) across Delhi NCR.

2nd Place, Spiritual Play Competition, TOVP Mayapur (July 2024).